Bad Altitude: Categorizing Elevation Produces Spurious Association With Concussions in the National Football League (NFL)

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Abstract

Objective: To assess whether prior analyses, where there was a relationship between altitude and concussion rates in American football, would replicate using a larger data set and altitude as a continuous variable.

Design: Cohort study replication

Methods: We analyzed data from all NFL regular season games from 2012-19. Concussions were identified from public databases and NFL injury reports. The altitude of each stadium was identified using mapping software. Concussion rates were calculated for each stadium and plotted against continuous altitude. We calculated crude rate ratios for several categorical cutpoints and used logistic and Poisson regression models to assess associations with continuous altitudes.

Results: We identified 867 players (1,103 player-seasons) who sustained 1,159 concussions during the time period 2012 to 2019. All continuous plots and models showed no evidence of any association between concussions and altitude. A Poisson model found an IRR of 1.00 (95% CI 0.99-1.01) for every 100 ft increase in altitude. A 644 ft cutpoint (used in previous studies) produced a significant difference (incidence rate ratio (IRR) 0.71, 95% CI: 0.54-0.94) in 2012-13, but this did not replicate in 2014-19 (0.99, 95% CI: 0.84-1.14).

Conclusions: We found no association between altitude and concussion rates in the NFL when altitude was analyzed continuously rather than inappropriately categorized. Our findings should increase skepticism of any effect of altitude on concussions at the elevations at which most American football is played as well as clinical interventions based on that theory. It also underscores the importance of keeping continuous variables continuous wherever possible.

Keywords: altitude, concussions, American football, statistics, categorization, continuous variables

Introduction

Previous research has suggested a lower risk for concussion risk when contact sports, including American football, are played at higher altitudes[9, 19, 27]. Exposure to extreme altitudes causes hypobaric hypoxia, which leads to a number of physiological responses. When altitude is sufficiently high to cause substantial hypoxemia (i.e., arterial partial pressure of oxygen less than 60mmHg) a number of physiological changes can occur in the brain, including increased cerebral blood flow. In some cases, the subsequent physiological changes in the brain (e.g., increased intracranial pressure) lead to altitude sickness or the life-threatening condition known as high altitude cerebral edema[5, 16, 19, 26, 27, 32, 34]. Some have suggested these same effects could be protective for concussion and traumatic brain injury, insofar as greater intracranial pressure may allow less "sloshing" of the brain during accelerations and decelerations[19, 27].

This "tighter fit" hypothesis of brain protection garnered early justification through epidemiologic investigations finding lower rates of concussion in high school American football and National Football League (NFL) games played at altitudes above 644 feet (196 m)[9, 19, 27]. However, other studies have found conflicting results: either no dose-response relationship between altitude and concussion rates when using quartiles of altitude or a higher concussion rate in college football games played at altitudes above 584 feet (178 m) or 932 feet (284 m)[17-19].

Others have argued the proposed physiological mechanisms are implausible at the low altitudes considered in these studiesat which NFL American football is regularly played[18, 29]. If a threshold for such a protective effect exists, it would likely only occur at altitudes that lower PaO_2 below the critical 60mmHg threshold:[16] an altitude greater than 8,202 ft (2500m), far beyond that of any NFL stadiums.

More importantly, previous studies suffer from a substantial statistical flaw: the inappropriate categorization of altitude – a continuous variable – for their primary analysis. Creating categories of altitude requires two questionable assumptions for valid results. First, one must assume no substantial variation of altitude effects within each category; for example, all stadiums from 645 ft (197m) above sea level to 5,194 ft (1,583m) above sea level should have similar concussion rates. Second, one must assume differences around the cutpoints; for example, stadiums 643 ft (196m) and 645 ft (197m) above sea level should have meaningfully different concussion rates. It is more likely that any altitude effects would occur continuously, though perhaps nonlinearly, as one moves from lower to higher altitudes. Furthermore, other studies have shown choosing irrelevant arbitrary cutpoints, such as animal vs. non-animal team logos, can yield statistically significant concussion results[28].

The objective of this study was to assess (i) whether prior analyses of altitude and concussions in the NFL would replicate using a larger data set and a continuous measure of altitude, and (ii) how robust the results were to minor changes in altitude cutpoints. We hypothesized that continuous analyses of altitude would show no association with concussion rates, and that prior categorical analyses would not replicate in future seasons.

Methods

We sought to replicate and extend earlier analyses by analyzing the correlation between stadium altitude and the rates of in-game concussions in the NFL.

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Data

Concussion data were collected from two sources: the PBS Concussion Watch blog (2012-15 seasons)[1] and an injury database kept by the analytics website Football Outsiders (FO, 2012-19 seasons)[8]. Both of these databases are based off public NFL injury reports issued by each team prior to their weekly games during the regular season, which list injured players, the type of injury they suffered including a separate category for concussions, their degree of participation in practices, and their likelihood of playing that week. We supplemented the injury data with additional manual searches for other concussions, or details about the concussions listed on the injury report, by Concussion Watch journalists or Football Outsiders staffers including the lead author of this paper. These lists are not comprehensive as not every concussion is reported publicly if it does not impact a player's participation at practice or in games either due to severity or timing (for example, occurring and resolving during a bye week). When comparing our concussion numbers to those officially reported by the NFL for 2015-19, we appeared to be missing 82 of 834 (10%) total concussions[20].

Concussions in the NFL are diagnosed by a team physician, on game days in concert with an unaffiliated neurotrauma consultant and/or athletic trainer. Diagnoses are made based on a combination of concussion signs and symptoms, video review of the injury, a focused neurological exam, and neurocognitive testing. The NFL has published its protocol publicly[21].

Data Management

The two sources were combined and cross-referenced to identify errors and eliminate duplicate entries for 2012-15; 2016-19 was solely from FO. Injuries in the FO database are collected prospectively by analysts and is based on the official public injury reports released weekly by each NFL team, supplemented with additional details from media reports where available. Information for all injuries included player name, team, position, week, season, final practice report status, and the player's participation in that week's game. Only concussions from weeks 1-16 of the regular season were included, as pre-season concussions were not reliably tracked, and non-playoff teams do not need to report injuries sustained in week 17. We could not differentiate between concussions that occurred in practices or in games, but per official NFL Health and Wellness reports, 95% (790/834) of regular season concussions from 2015-19 occurred in games[20].

Altitude was collected by first identifying the latitude and longitude of each NFL stadium on Wikipedia and Google Maps; the chosen coordinates represented approximately the center of each stadium's playing field. These coordinates were then input into an elevation map tool from the United States Geological Survey[30] for U.S. stadiums or an alternative elevation map tool[10] for international sites to get a measure of elevation above sea level to the nearest foot. These altitudes are listed in **TABLE 1**.

Game location was typically the home team's stadium, with the exception of two games moved to Detroit due to weather (Buffalo Bills vs. New York Jets, 2014), as well as NFL International Series games in the UK, Mexico City, and Toronto, Canada. All non-traditional stadium locations (i.e. international games and games played at neutral sites due to weather) and between-season stadium moves during the study timeframe (e.g. the Rams moving to Los Angeles in 2016 or the Falcons switching stadiums in 2017) were accounted for. The altitude was always recorded for the exact venue in which the games were played.

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Statistical Analysis

First, we counted the total number of concussions by year to ensure we had a similar dataset to prior studies[9, 19]. We then calculated the rate of concussions per game by stadium from 2012-2019. We plotted these rates against the log of altitude for 2012-13 (the timeframe of Myer et al), 2012-15 (the timeframe of Connolly et al), 2014-2019 (the 6 seasons after Myer et al), and 2012-2019 (all time periods together), with the predictions of a generalized additive model (GAM) with a restricted cubic spline basis function and altitude as its single predictor overlaid. GAMs automatically fit penalized smooth curvilinear associations to the data, allowing one to detect non-linear relationships while avoiding over-fitting[14]. For this portion of the analysis, we excluded stadiums with fewer than 4 games in each time period: Toronto, Mexico City, and non-Wembley English stadiums.

We then calculated crude concussion rates per game and replicated and tested the robustness of earlier results by shifting the 644 ft cutoff up or down by 100 ft (30m) for the 2012-13 seasons; testing a true median split (≤260 ft (79m) vs. ≥397 ft (121m)) of NFL stadiums during that time period; and attempting to replicate all these categorical results in 6 future seasons (2014-19)[9, 19].

Finally, we organized the data by team-game such that there were 2 observations per game. To replicate and extend prior papers, we used two mixed effect regression methods to assess the association between continuous and categorical (644 ft (196m) cutpoint) altitude and the risk of 1+ concussions in a team-game (logistic) and the rate of concussions per team-game (Poisson). For the Poisson models we investigated but found no evidence of zero-inflation nor any substantial influence of outliers. Team-level random effects were used to account for intra-team correlation. We adjusted for whether each team was at home or away. We also adjusted for season and week in sensitivity analyses, but these models did not differ given the lack of any systematic relationship between these variables and altitude, so only the simpler model results are reported.

We conducted a sensitivity analysis with negative binomial models to account for possible overdispersion; those results were very similar to the Poisson models and are not reported separately. Based on the lack of any evident non-linear relationship in the plots, we assumed a linear relationship for the logistic and Poisson regressions. We conducted these analyses for 3 of the time periods described above: 2012-13, 2014-19, and 2012-2019.

All analyses were conducted in R version 4.0.2 with the tidyverse, lme4, epitools, patchwork, fuzzyjoin, and lubridate packages.[4, 6, 13, 22-24, 31] The code for the analyses reported in this manuscript as well as sensitivity analyses is available on Github (https://github.com/zbinney/altitude_concussions). All data are available in an interactive app that also allows replication of our analyses for different years and altitude cutpoints (https://zbinney.shinyapps.io/concussions_altitude_shinyapp/).

This study did not require IRB approval or participant informed consent as it did not meet the criteria for human subjects research, and all data were publicly available.

Results

We identified 1,159 concussions in 867 players (1,103 player-seasons) from 2012-19; there were 300 concussions in 269 players (285 player-seasons) in 2012-13, closely matching earlier analyses of that period[19].

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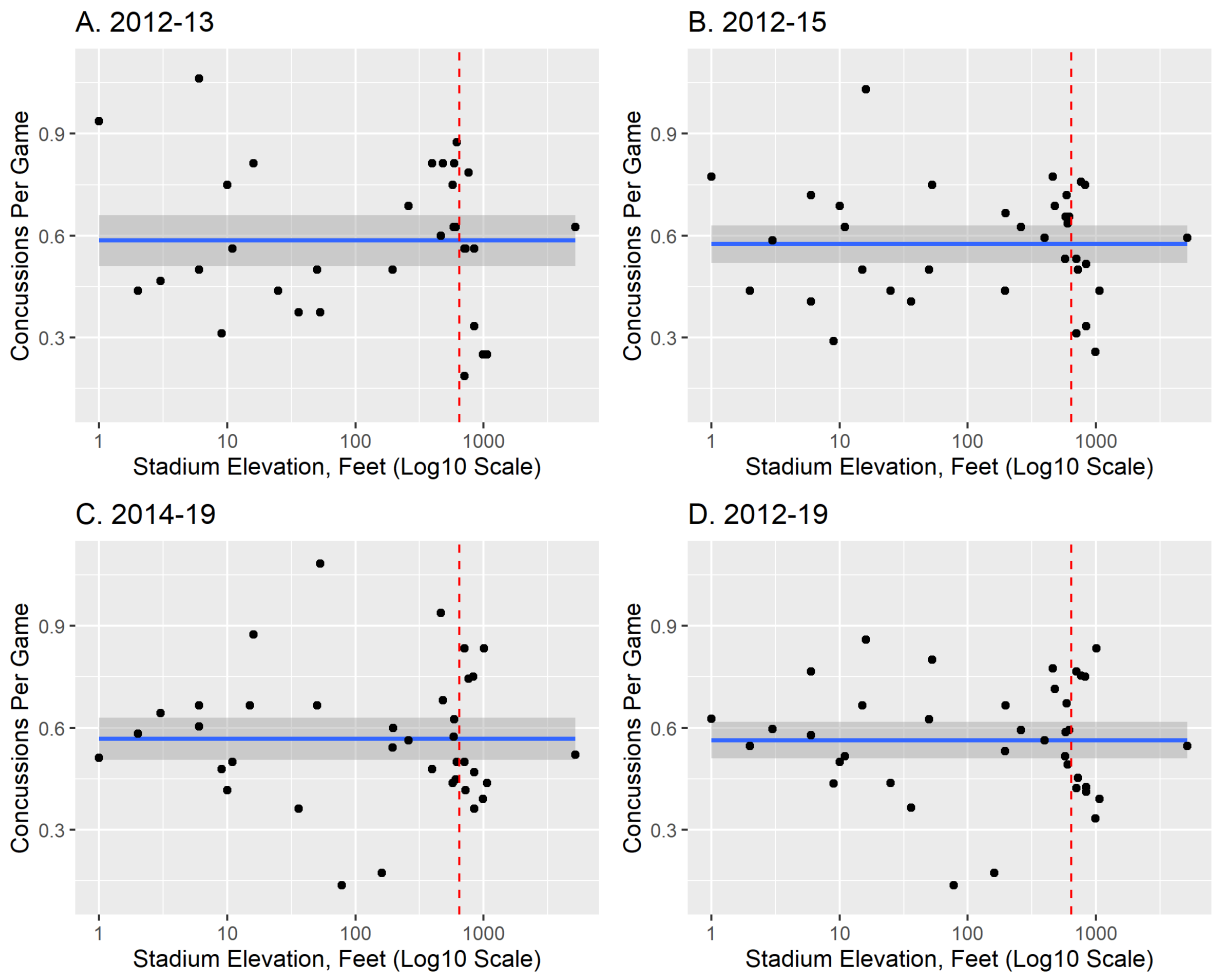
FIGURE 1 plots concussion rates vs. continuous altitude for stadiums with ≥4 games played during the indicated time period. The plots and GAMs suggested no association between altitude and concussion in any of these time periods. The vertical red lines show the 644 ft altitude cutpoint. Note all plots show >32 points due to international games and city or stadium moves during the indicated time periods. In all time periods Denver, the highest-altitude NFL stadium that regularly hosts games, has approximately average concussion rates. Mexico City's Estadio Azteca has an elevation of 7,503 feet but has hosted too few games to generate a reliable estimate of its concussion rate (0.33, 95% CI 0.01-1.86). Consistent with **FIGURE 1**, there was no association between continuous altitude and concussion rates or the odds of a team suffering 1+ concussions in a game (**TABLE 3)**.

FIGURE 1. Concussion Rate vs. Continuous Altitude in the NFL.



The solid lines on each plot are a fitted GAM; the shaded area is its 95% confidence interval.
The vertical dashed lines demarcate the 644 ft (196m) cutpoint used in prior literature.

TABLE 2 and **TABLE 3** show our replications of Myer et al's results for a 644 ft (196m) cutpoint and the 2012-13 seasons. We found nearly identical results, with a crude incidence rate ratio (IRR) of 0.71 (95% CI: 0.54-0.94) and an adjusted odds ratio of 0.67 (95% CI: 0.47-0.94). The first rows of **TABLE 2** and **TABLE 3** show these relationships disappeared in the 2014-19 time period. The 2012-13 column of

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TABLE 2 also shows the IRR was sensitive to the particular choice of cutpoint; our three alternatives are all attenuated towards a null IRR of 1.00, and none remained statistically significant.

Discussion

We found no association between altitude and concussion rates in the NFL when altitude was analyzed continuously rather than inappropriately categorized (**FIGURE 1** and **TABLE 3**). Previous results in this field are not robust to changes in the specific cutpoint chosen to generate altitude categories or repeatable in future years. While Myer et al found an IRR of 0.71 (95% CI: 0.54-0.94) for games occurring at ≥644 ft (196m) vs. <644 ft in 2012-13, our analysis yielded an IRR of 0.99 (95% CI: 0.84-1.14) in the subsequent 6 seasons. Moving the cutpoint up or down 100 feet also substantially attenuated the relationship, while using a median altitude split of NFL stadiums (397 ft or 121 m) eliminated the relationship entirely[27]. Our interactive app allows users to explore how different cutoff points and seasons change the IRR.

The 644 ft (196m) cutpoint originated in Smith et al's analysis of surveillance data for U.S. high school American football athletes[27]. The reason for 644 ft as the cutpoint was not explained, but it was implied to be the median altitude of schools that provided American football data to the surveillance system. Thus this threshold was based on arbitrary sample characteristics rather than a physiologic rationale, and it just happened to fall between one group of 8 stadiums with concussion rates above the median and another group of 8 stadiums where concussion rates were below the median during this time period (**FIGURE 1A**). The pattern still largely holds when Connolly et al added data in 2014 and 2015[9] (**FIGURE 1B**). However, in independent or longer time periods, this chance pattern isn't replicated, and earlier statistically significant findings disappear[9, 19] (**FIGURE 1C-D**).

This, combined with the lack of any association of concussion rates with continuous altitude, suggest categorical analyses indicating a relationship between higher altitude and lower concussion rates in American football are most likely statistical artifacts that do not represent causal effects of altitude.

Other Studies on Concussions and Altitude

The concussion field has a mix of studies claiming a protective effect of higher altitude, a protective effect of lower altitude, and neither[9, 17-19, 27] (**TABLE 4**). A meta-analysis of four studies, which included nearly 5 million athlete-exposures, concluded there was no consistent association between altitude and concussion incidence in American football[34]. Even after this and other papers calling epidemiologic data on altitude and concussions into question[28, 29, 33, 34], research continued in both American football and hockey using the same approach of categorizing altitude[3, 9]. We question the conclusions.

Implications for Clinicians, Practitioners, Coaches and Athletes

Practitioners, coaches, and athletes need to know that, contrary to some previous findings, concussions in American football do not occur less (or more) frequently at higher altitudes, and the epidemiologic evidence does not support the "sloshing" and "tighter fit" hypotheses. We urge caution when practitioners, coaches and athletes are considering altering concussion safety or rehabilitation practices based on current elevation or anticipating a protective effect by moving training or game play to modestly higher altitudes.

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The statistically significant association between concussion rates and altitude have previously been used to support the hypothesized mechanism underlying a jugular compression collar that claims to mimic the protective effects of playing at altitude[2, 19]. We urge clinicians, coaches, strength and conditioning staff, and equipment managers who are considering using these devices to pay attention to new negative findings when choosing to purchase or use the collars in the hope of protecting against brain injuries.

Implications for Researchers

Clinicians and practitioners should know that the ill-advised categorization of continuous variables is common, and even the default, in much of the medical literature[7, 12]. Connolly et al begin their results section with two plots of concussion rate per game by continuous altitude that mimic our analysis, but proceed to ignore them in favor of other analyses using the 644 ft cutpoint.[9] Our analysis highlights how such categorizations can lead to incorrect conclusions.

We hope readers are now sufficiently prepared to watch for this same error in other studies. Instead of categorizing continuous variables, we encourage researchers to use techniques such as linear regression or generalized linear models (e.g. Poisson or logistic regression) when the relationship between the continuous variable and outcome of interest is likely to be linear[7]. If the relationship is non-linear, rather than categorical binning (i.e. categorizing) we encourage researchers to consider splines, which split associations into multiple line segments, or generalized additive models, which fit arbitrary smooth curvilinear relationships[7, 11, 15, 25].

Limitations

We only had access to data on publicly-reported concussions. When comparing our concussion numbers to those officially reported by the NFL for 2015-19, we appeared to be missing 82 of 834 (10%) total concussions[20]. This is likely because not every concussion tracked in the NFL's Injury Surveillance System is listed in public injury reports, but there is little reason to think this biases our results, as missingness should not be systematically associated with altitude. Second, we were also unable to differentiate when multiple concussions that occurred in close succession were separate incidents or aggravations of prior injuries. Third, we were unable to identify whether a concussion occurred in a practice or a game. Although, this is unlikely to bias our results as 95% of regular season concussions occur in games, and a majority of those occurring in practices would have occurred at or near the altitude of the stadium in which that week's game was played[20]. Fourth, we did not have data on college or high school American football for comparison.

Conclusion

There was no relationship between altitude and concussions in the NFL from 2012-19. No further research is needed on this topic and further epidemiologic investigations are unlikely to be fruitful. Our findings should increase skepticism of any effect of altitude on concussions at the elevations at which most American football is played. It also underscores the importance of keeping continuous variables continuous wherever possible and the dangers of finding spurious associations – or potentially obscuring or biasing true associations – when they are categorized.

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Key points

- **Findings:** There is no association between altitude and concussion rates in professional American football when altitude is considered continuously
- **Implications:** Previous relationships between altitude and concussions do not replicate when using improved statistical methods. Practitioners and athletes should use caution when considering altering concussion safety or rehabilitation practices based on elevation or anticipating elevation changes to reduce concussion risk. Continuous variables such as altitude should only be categorized if everyone in a single category can be considered similar and those near the category cutoffs can be considered different
- **Caution**: These findings may not generalize to other sports or settings.

Study Details

Data Sharing: All data referenced in the manuscript is available for download and interrogation via an interactive app here.

Ethics approval: This study used only publicly available data and as such was exempt from ethics review.

Contributions: ZB and JS conceived of the idea for the paper, designed the study, and wrote the manuscript. ZB was responsible for data collection and analysis. All authors read and approved the final manuscript.

Patient and Public Involvement in Research: No patients, athletes, or public partners were involved in this research.

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TABLE 1. List of stadiums and altitudes.

Stadium	Elevation (Feet)
Arizona	1068
Atlanta (2012-16)	989
Atlanta (2017-19)	1009
Baltimore	10
Buffalo	763
Carolina	707
Chicago	592
Cincinnati	480
Cleveland	582
Dallas	575
Denver	5195
Detroit	604
Green Bay	620
Houston	50
Indianapolis	708
Jacksonville	3
Kansas City	842
Los Angeles Chargers (2017-19)	78
Los Angeles Rams (2016-19)	161
London (Wembley)	62
London (Twickenham)	85
London (Tottenham)	197
Mexico City	7503
Miami	9
Minnesota (2014-15)	827
Minnesota (2012-13, 2016-19)	844
New England	260
New Orleans	2
New York	6
Oakland	0
Philadelphia	11
Pittsburgh	725
San Diego (2012-16)	53
Seattle	16
San Francisco (2014-19)	15
San Francisco (2012-13)	25
St. Louis Rams (2012-15)	463

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Tampa Bay	36
Tennessee	397
Toronto	397
Washington, DC	195

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TABLE 2. Crude Concussion Rates and Rate Ratios Per Game by Categorical Altitude Cutpoints for 2012-13, 2014-19, and 2012-19.

	2012-13		2014-19		2012-19	
	Concussion Rate	**IRR**	**Concussion Rate**	**IRR**	**Concussion Rate**	**IRR**
644 ft Split						
>=644 ft	0.45	0.71 (0.54-0.94)	0.55	0.99 (0.84-1.14)	0.53	0.91 (0.80-1.04)
<644 ft	0.64	--	0.56	--	0.58	--
744 ft Split						
>=744 ft	0.46	0.76 (0.54-1.03)	0.54	0.96 (0.80-1.13)	0.52	0.90 (0.77-1.05)
<744 ft	0.61	--	0.56	--	0.58	--
544 ft Split						
>=544 ft	0.56	0.92 (0.73-1.15)	0.54	0.94 (0.82-1.08)	0.54	0.94 (0.83-1.06)
<544 ft	0.61	--	0.57	--	0.58	--
NFL Median Split						
>= 397 ft	0.59	1.00 (0.79-1.25)	0.55	0.98 (0.86-1.12)	0.56	0.98 (0.88-1.11)
<397 ft	0.59	--	0.56	--	0.57	--

Abbreviations: IRR, (incidence) rate ratio. All IRRs presented as Point Estimate (95% confidence interval)

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TABLE 3. Logistic and Poisson Generalized Linear Mixed Models for Association Between Concussions and Altitude.

	2012-13		2014-19		2012-19	
	Logistic OR	**Poisson IRR**	**Logistic OR**	**Poisson IRR**	**Logistic OR**	**Poisson IRR**
644 ft Split						
>=644 ft	0.67 (0.47-0.94)	0.72 (0.54-0.96)	1.01 (0.83-1.23)	1.00 (0.86-1.18)	0.91 (0.77-1.08)	0.93 (0.81-1.07)
<644 ft	--	--	--	--	--	--
Continuous Altitude						
100' increase	1.00 (0.98-1.01)	1.00 (0.99-1.01)	1.00 (0.99-1.01)	1.00 (0.99-1.01)	1.00 (0.99-1.01)	1.00 (0.99-1.00)

Abbreviations: IRR, (incidence) rate ratio; OR, odds ratio. All numbers presented as Point Estimate (95% confidence interval).

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TABLE 4. Review of Prior Studies on Altitude and Concussions in American football.

Study	Population	Altitude Categorization	Results	Results Favor
Smith et al 2013[27]	High School Athletes, 2005-12	<644 ft (196m) vs. ≥644 ft (196m)	IRR, using AEs: 1.28 (95% CI: 1.18-1.37)	"Higher" Altitude
Myer et al 2014[19]	NFL, 2012-13	<644 ft (196m, N = 24) vs. ≥644 ft (196m, N = 9)	IRR using AEs: 0.70 (95% CI: 0.53-0.94) Adjusted OR for 1+ concussions in athlete in a game: 0.68 (95% CI: 0.47-0.99)	"Higher" Altitude
		Quartiles: 0-19 ft (0-6m, N = 9), 20-399 ft (6-122m, N = 8), 400-709 ft (122-216m, N = 8), ≥710 ft (≥216m, N = 8)	Adjusted OR for 1+ concussions in athlete in a game: 1st quartile ref., 2nd quartile 0.93 (0.62-1.40), 3rd quartile 1.02 (0.71-1.45), 4th quartile 0.67 (0.43-1.05)	
Lawrence et al 2016[17]	NFL, 2012-13	Quartiles: ≤17 ft (≤5.2m), 19-397 ft (5.9-121.0m), 466-710 ft (142.0-216.3m), ≥724 ft (≥220.6m)	IRR, using games, estimated from plot: 1st quartile ref., 2nd quartile ≈0.9 (0.7-1.3), 3rd quartile ≈1.1 (0.8-1.5), 4th quartile ≈0.8 (0.5-1.1)	Neither
Lynall et al 2016[18]	21 NCAA Division I American football teams playing in 168 venues, 2009-13	Median: >584 ft (>178m) vs. <584 ft (<178m)	IRR, using AEs: 1.47 (95% CI: 1.09-2.00)	"Lower" Altitude
		Quartiles: <141 ft (<43m), 180-574 ft (55-175m), 597-932 ft (182-284m), >932 ft (>284m)	IRR, using AEs: 1st quartile ref.; 2nd quartile 0.90 (95% CI: 0.57-1.41); 3rd quartile 0.91 (95% CI: 0.54-1.53); 4th quartile 1.67 (95% CI: 1.13-2.48)	
Connolly et al 2018[9]	NFL, 2012-15	≥644 ft (196m, N = 9) vs. <644 ft (196m, N = 23)	IRR, using games, calculated by us: 0.78 (95% CI: 0.64-0.94). NOTE: authors say they did not find a statistically significant relationship at this cutpoint.	"Higher" Altitude
		Quartiles: 0-19 ft (0-6m, N = 8), 20-397 ft (6-121m, N = 8), 398-704 ft (121-215m, N = 8), 705-5280 ft (215-1,609m, N = 8)	Concussion rate per game, estimated from plot: 1st quartile ≈0.6 (95% CI: 0.5-0.8), 2nd quartile ≈0.6 (95% CI: 0.5-0.7), 3rd quartile ≈0.7 (95% CI: 0.6-0.8), 4th quartile ≈0.5 (95% CI: 0.4-0.6)	

Abbreviations: IRR = (incidence) rate ratio; OR = odds ratio; AE = athlete exposure